

- 1 *Chlorella* is a green, single-celled protist that is photosynthetic. Some students immobilised cells from a culture of this protist using the same method as used for immobilising enzymes.

(a) Outline how the students could immobilise cells of *Chlorella*.

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The students used the respirometer shown in Fig. 1.1 to measure oxygen uptake by the immobilised *Chlorella* cells.

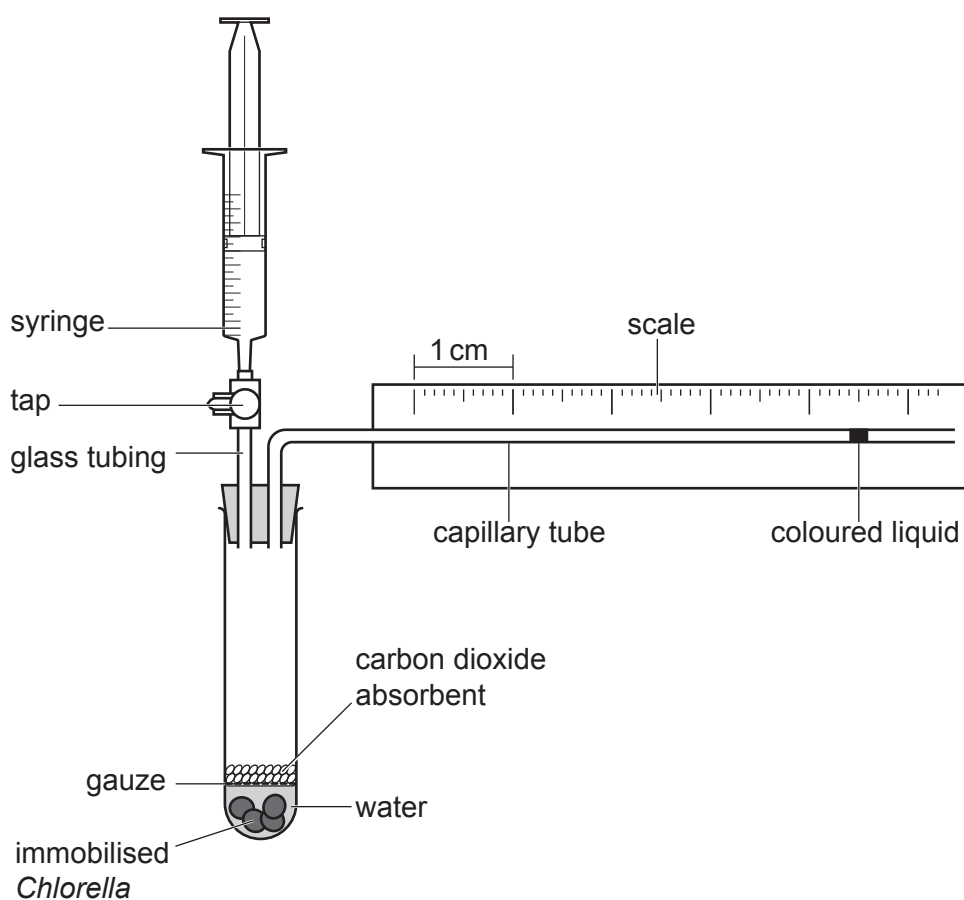


Fig. 1.1

The students used the respirometer to test the hypothesis:

The volume of oxygen taken up by respiration in immobilised *Chlorella* increases with an increase in temperature.

(b) (i) State the independent variable and the dependent variable in this investigation.

independent variable

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dependent variable

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[2]

(ii) The students decided to measure oxygen uptake for 5 minutes using a range of temperatures from 10°C to 50°C.

Discuss whether the range of temperatures used by the students was suitable for their investigation.

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- (c) The students calculated the oxygen uptake for each temperature.

Their results are shown in Table 1.1.

Table 1.1

temperature / °C	10	20	30	40	50
oxygen uptake / mm ³	200	400	800	750	680

- (i) Outline how the students calculated the volume of oxygen uptake.

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- (ii) State **and** explain **two** conclusions about the trends shown by these results.

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[Total: 18]